

SCHEMA AND DATABASE REPORT

ChallengeON DCT Analytics Lake

Physical schemas, semantic views, lineage, validation results, and operational query guidance.

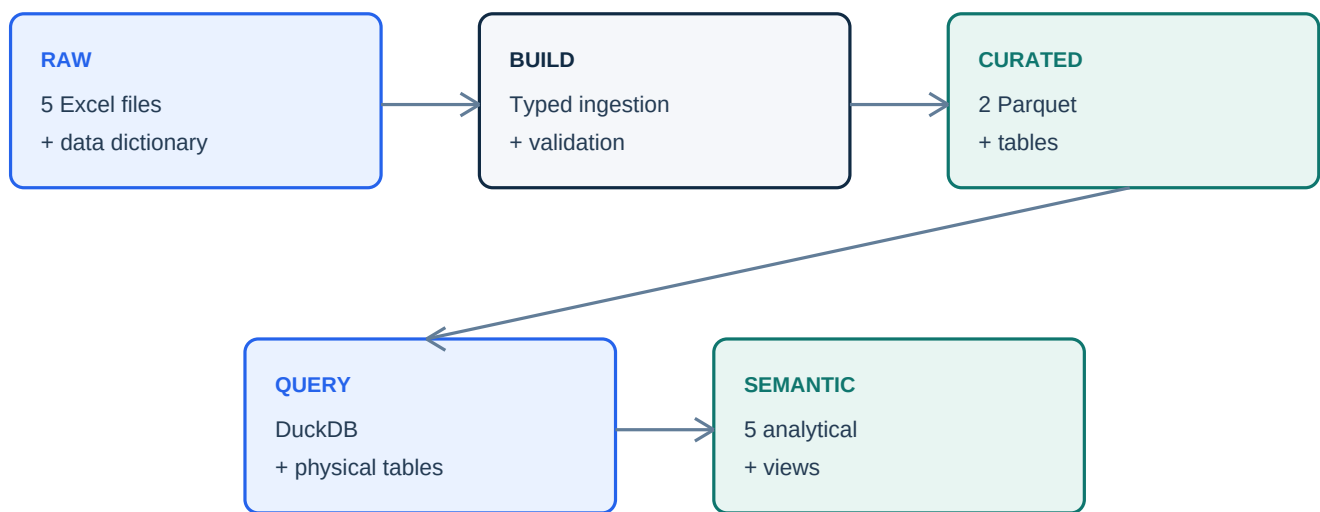
DATABASE	analytics.duckdb
DATA WINDOW	01 Jan 2022 - 28 Feb 2026
REPORT DATE	28 Sep 2026

01 Executive summary

186,952 Rows across physical tables	2 Curated physical tables	5 Analytical views	2.0 MB Curated Parquet footprint
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The database is a compact analytical layer over two canonical Parquet datasets. The physical tables retain source lineage and nullable source fields; semantic views separate training observations, prediction rows, daily aggregations, and the safe guest-to-flight join.

Lake architecture



DESIGN NOTE

Parquet is the portable source of truth. DuckDB is a generated query artifact that can be rebuilt from the raw workbooks using `scripts/build_lake.py`.

Database inventory

Object	Kind	Rows / role
guest_daily	Physical table	69,344 guest observations and prediction rows
flight_daily	Physical table	117,608 route-airline-date observations
5 semantic views	Views	Target filtering, daily aggregation, and safe fact join

02 Physical schema: guest_daily

Grain: one row per date and residence group, with nationality included for International records. Domestic nationality is NULL by design. Candidate key: date + residence_group + nationality.

69,344 Rows	59,930 Training targets	9,414 Prediction rows	45 Nationalities
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Column	Type	Nullable	Definition
date	DATE	Yes	Calendar date for the guest observation.
residence_group	VARCHAR	Yes	Domestic or International segment.
nationality	VARCHAR	Yes	Guest nationality; NULL for domestic rows.
guests	BIGINT	Yes	Prediction target. NULL in the test/prediction split.
new_arrivals	BIGINT	Yes	Guests newly arriving on the date.
same_day_guests	BIGINT	Yes	Same-day visitors. NULL can mean suppressed, unavailable, or not applicable.
dataset_split	VARCHAR	Yes	Train or test source classification.
source_file	VARCHAR	Yes	Original workbook filename for lineage.

Segment and split profile

Residence	Split	Rows	Target rows	Prediction rows
Domestic	test	212	0	212
Domestic	train	1,308	1,308	0
International	test	9,202	0	9,202
International	train	58,622	58,622	0

DATA NOTE

same_day_guests contains 39,428 NULL values (56.9% of rows). These must not be blindly converted to zero because the source dictionary combines suppressed, unavailable, not-applicable, and zero-like meanings.

03 Physical schema: flight_daily (1 of 2)

Grain: one row per date, origin country, origin city, arrival city, airline, and destination. The current candidate key is unique across all 117,608 rows.

117,608 Rows	1,167 Distinct dates	33 Countries	59 Airlines
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Column	Type	Nullable	Definition
date	DATE	Yes	Operating date represented by the record.
departure_country_name	VARCHAR	Yes	Country from which the flight departs.
departure_city	VARCHAR	Yes	Origin city.
arrival_city	VARCHAR	Yes	Arrival city; currently Abu Dhabi.
airline_name	VARCHAR	Yes	Operating carrier name.
average_weekly_frequency	DOUBLE	Yes	Average weekly service frequency where supplied.
business_class_p2p_count	BIGINT	Yes	Point-to-point Business Class passengers.
business_class_seat_capacity	BIGINT	Yes	Business Class seats offered.
economy_class_p2p_count	BIGINT	Yes	Point-to-point Economy Class passengers.
economy_class_seat_capacity	BIGINT	Yes	Economy Class seats offered.
first_class_p2p_count	BIGINT	Yes	Point-to-point First Class passengers.
first_class_seat_capacity	BIGINT	Yes	First Class seats offered.
load_factor	DOUBLE	Yes	Total passengers divided by total seats.

DESIGN NOTE

load_factor is stored as a ratio, not a percentage string. Validation confirms $\text{load_factor} = \text{total_pax} / \text{total_seats}$ within floating-point tolerance.

04 Physical schema: flight_daily (2 of 2)

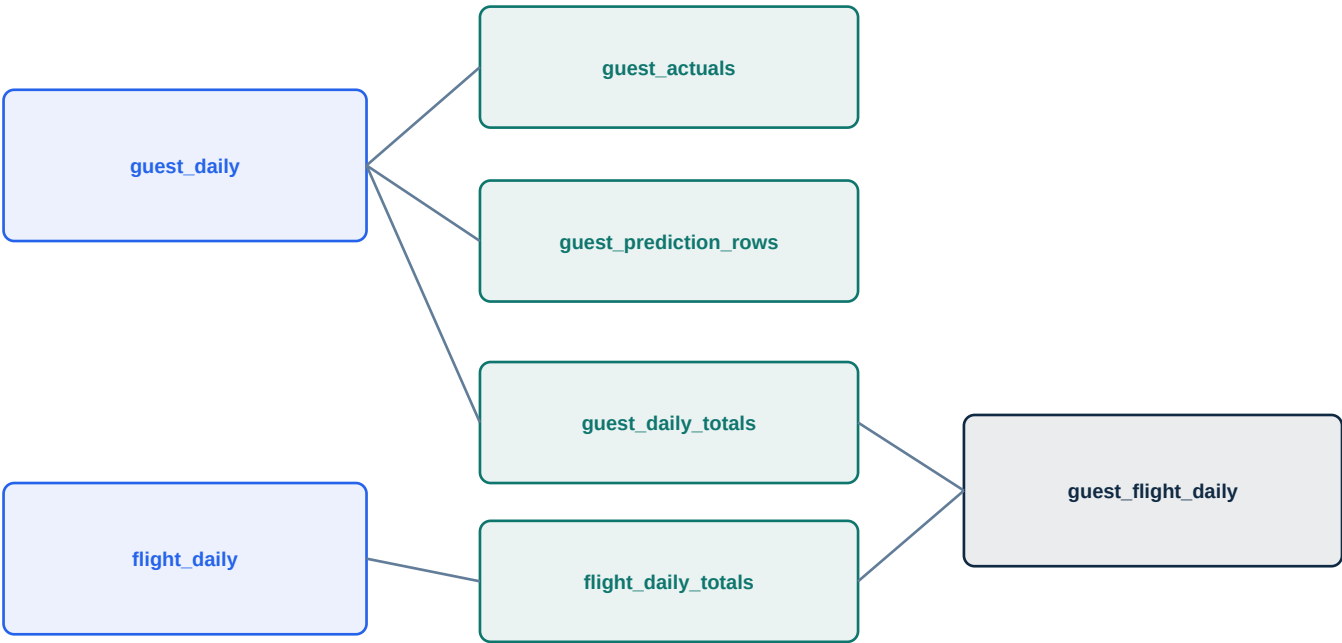
Column	Type	Nullable	Definition
total_p2p	BIGINT	Yes	All point-to-point passengers.
total_pax	BIGINT	Yes	All passengers, including P2P, transfer, and transit.
total_pax_excluding_infant	BIGINT	Yes	Passengers excluding infants.
total_seats	BIGINT	Yes	Total seat capacity.
total_transfer	BIGINT	Yes	Passengers transferring to another flight at Abu Dhabi.
total_transit	BIGINT	Yes	Passengers transiting through Abu Dhabi.
transfer_business_class_count	BIGINT	Yes	Transfer passengers in Business Class.
transfer_economy_class_count	BIGINT	Yes	Transfer passengers in Economy Class.
transfer_first_class_count	BIGINT	Yes	Transfer passengers in First Class.
transit_business_class_count	BIGINT	Yes	Transit passengers in Business Class.
transit_economy_class_count	BIGINT	Yes	Transit passengers in Economy Class.
transit_first_count	BIGINT	Yes	Transit passengers in First Class.
destination	VARCHAR	Yes	Destination IATA code; currently AUH.
source_file	VARCHAR	Yes	Original workbook filename for lineage.

Selected null profile

Column	NULL rows	Share	Interpretation
average_weekly_frequency	1,213	1.03%	Unavailable for a small initial block
business_class_p2p_count	1,213	1.03%	Suppressed or unavailable
economy_class_p2p_count	1,213	1.03%	Suppressed or unavailable
first_class_p2p_count	1,213	1.03%	Often unavailable or not applicable
total_pax_excluding_infant	1	0.001%	One missing record

05 Semantic views and lineage

The semantic layer prevents accidental fact-to-fact multiplication. Guest and flight facts are each aggregated to date before the combined view joins them.



View dictionary

View	Purpose	Primary input
flight_daily_totals	Flight facts aggregated to daily demand, capacity, and weighted load factor.	flight_daily
guest_actuals	Guest rows with a populated target for analysis and model training.	guest_daily
guest_daily_totals	Guest facts aggregated to one row per date and dataset split.	guest_daily
guest_flight_daily	Safe daily join between guest totals and flight totals.	daily aggregate views
guest_prediction_rows	Guest rows whose target must be predicted.	guest_daily

DESIGN NOTE

DuckDB does not enforce the documented candidate keys. The lake build validates uniqueness before publishing the curated tables and records the results in manifest.json.

06 Validation results and known caveats

Validation	Result	Rule	Status
Guest candidate-key duplicates	0	Must equal 0	PASS
Flight candidate-key duplicates	0	Must equal 0	PASS
Training rows missing target	0	Must equal 0	PASS
Test rows with populated target	0	Must equal 0	PASS
Passenger identity mismatches	0	total_pax = p2p + transfer + transit	PASS
Maximum load-factor error	1.11e-16	Tolerance <= 1e-9	PASS

Known caveats

DATA NOTE

Flight grain discrepancy: the supplied data dictionary describes monthly records, but the workbook contains daily dates from 2023 onward. The implemented schema follows the observed workbook grain.

DESIGN NOTE

Target availability is intentional: all 9,414 test rows have guests = NULL, while all 59,930 training rows have a populated target.

DESIGN NOTE

Arrival city and destination are currently constant (Abu Dhabi and AUH). They remain in the curated table to preserve lineage and support future multi-destination extracts.

Source traceability

manifest.json stores byte sizes and SHA-256 hashes for every Excel source, the data dictionary, row-count checks, and the two curated output paths. This makes rebuilds auditable without modifying the raw source files.

07 Query and operating guide

Rebuild the lake

```
source .venv/bin/activate
python scripts/build_lake.py
```

Query the safe daily guest-flight view

```
.venv/bin/python scripts/query_lake.py \
"SELECT date, guests, total_pax, flight_load_factor
FROM guest_flight_daily
ORDER BY date DESC LIMIT 10"
```

International training history

```
SELECT date, nationality, guests, new_arrivals
FROM guest_actuals
WHERE nationality = 'INDIA'
ORDER BY date;
```

Monthly flight demand

```
SELECT date_trunc('month', date) AS month,
       SUM(total_pax) AS total_pax,
       SUM(total_seats) AS total_seats,
       SUM(total_pax)::DOUBLE / NULLIF(SUM(total_seats), 0) AS load_factor
FROM flight_daily
GROUP BY month
ORDER BY month;
```

Operating principles

Principle	Implementation
Raw immutability	The original workbooks and PDF remain unchanged under 01a - DCT Dataset.
Reproducible publication	The builder replaces generated Parquet and DuckDB outputs only after validation succeeds.
Portable storage	Parquet files are the versionable curated layer; DuckDB is generated locally.
Safe joins	Use guest_flight_daily unless route-level flight detail is explicitly required.
NULL preservation	Suppressed and unavailable source values stay NULL rather than becoming plausible zeros.

Report sources

analytics.duckdb; manifest.json; Data_Dictionary.pdf; data domestic_train.xlsx; data domestic_test.xlsx; data international_train.xlsx; data international_test.xlsx; flight_data.xlsx.